

Team Performance Prediction on Team Sports: A Complex Network Approach

Zhongyuan Yu, Hanchao Yang

October 2017

Abstract

In this paper, we introduce a model mapping individual players performance to a team complex network. The network reveals their collaborative relations as a team. The team performance is measured and predicted by network efficiency. An empirical analysis is given on basketball teams from U.S. National Basketball Association (NBA). It is showed that team network efficiency and winning probability is positively and linearly correlated.

Keywords: team performance, complex network, network efficiency, basketball

1 Introduction

In this paper, complex network is proposed to discover the relation between individual player statistics and team performance. We construct networks for each NBA team based on player box statistic data while players are connected nodes. Such network topological features reveal cooperative relationships among players. A "hub" in complex network is a node highly connected to others. It is showed from the team network that important players are well connected in basketball teams. Further, we estimate network efficiency as a tolerance of player failure or underperformance. It is found empirically that there exists a positive linear relation between network efficiency and winning probability.

We refer error tolerance to the ability of remain team collaborative when one or a number of players suffer from risk events such as foul-out, injury, suspension and extreme underperformance. For example, a portion of players will suffer from extreme underperforming if their opponent team developed a very successful defensive strategy against them. In a team network, extreme underperformed players are recongized as disfunctioned or removed nodes. Since team sports are highly dynamic, the ability of remain collaboration and keep other players perform well is directly related to the team performance. In network theory, a removal of random nodes is defined as an error. Error tolerance is robustness of a network structure and is measured by network

efficiency (Latora and Marchiori, 2001; Alber et al, 2000). We will show in empirical analysis that network efficiency is positively correlated with winning probability.

The paper is related to several literatures on different topics. Vaz de Melo et al. (2008) analyzed the evolution of the NBA using complex network metrics, calculated from a graph in which the vertices are the players and the teams, and the edges identify labor relationships among them. It was shown that NBA network structure can be characterized as a small-world network and complex network metrics provide good prediction information without using box score statistics. Park and Newman (2005) introduced a ranking system for division I-A American college football based on networks and linear algebra. Vaz de Melo et al. (2012) used the network-based prediction models, which can indicate how well a team will perform over a season.

Basketball sport is highly dynamic so that to accurately capture the whole performance of a player and a team by numbers is very challenging. Statistics such as Points Per Game (PPG), Assists Per Game (APG) are treated separately and the players are treated separately so that the hidden relations are ignored. In history, there are players with insignificant box scores statistics playing significant roles on a team success (Vaz de Melo et al. 2008). Moreover, since NBA announced several rules and regulations to balance the development of league teams such as draft rules, salary caps etc., the structuring a team is more important and practical than simply hiring all-stars. Teamwork plays a crucial role deciding wins or loses much more than individual performance.

2 Methodology

2.1 Construction a Team Network

Firstly, we define a sport team as a network $G(V, E, \mathcal{W})$ where a node or vertex represents a player $v_i \in V$. A node or a player is described by a set of information $v_i = \{I_1, I_2, \dots, I_n\}$. Information set $I = \{I_1, I_2, \dots, I_l\}$ is referred to values such as height, weight, field goal, pass, speed etc. E is edges between two nodes while $\mathcal{W} : v_i, v_j \rightarrow E_{i,j}$ is a weight function of the edge between v_i and v_j .

Then, an utility function of a player is defined from his information set to utility score. Notice that the utility score varies due to the purpose of team management. It could contain information like salaries, injuries, insurance costs, etc. Or it could only consider offensive performance such as offensive rebounds, shoots, batting average, etc. Here we show a general definition of utility score. A specific representation for basketball teams can be found in definition (10).

Definition 1 (Utility Score). *Let $I = \{I_1, I_2, \dots, I_l\}$ is the information set of a*

player. Then we have a function $U : I \rightarrow S$ where S is the utility score.

$$s_i = f(I_1, I_2, \dots, I_l), \forall v_i \in V \quad (1)$$

In our view, a player with high utility score should have a high degree. A degree is number of connections a node has to the other nodes. Degree distribution is a probability distribution of degrees of network. In sports, a key player in a team is always the core in many tactical strategies so that his collaboration with other team members is more important than bench players. In other words, he's more connected to others and has a higher degree.

We use power law mapping utility score to degrees sequence such that:

Definition 2 (Degree).

$$D(v_i) = s_i^\lambda, \forall v_i \in V \quad (2)$$

where $D(v_i)$ is the degree of node v_i . λ is a parameter whose value is usually in the range $2 < \lambda < 3$ for real-world networks (Onnela et al. 2007).

With degree sequence defined in (2), our network structure or adjacency matrix A is generated from Havel-Hakimi algorithm (Havel 1955; Hakimi 1962). It is an algorithm in solving graph realization problem that create an undirected unweighted network with given degree sequence. Nodes with high degree will be linked with priority. It is in accordance with sports that core players have higher priority to cooperate with each other than it with assistant players.

Adjacency matrix is defined as:

Definition 3 (Adjacency matrix).

$$a_{i,j} = \begin{cases} 1 & \forall \{v_i, v_j\} \in E, i \neq j \text{ (connected)} \\ 0 & \text{otherwise} \end{cases} \quad (3)$$

A is actually a matrix representation of the edge set E . A is symmetric since G is undirected.

In a sport team, we model a team as a weighted network for the importance of measuring level of collaboration.

Definition 4 (Weight Function). We define a weight function $\mathcal{W} : v_i, v_j \rightarrow E_{i,j}$:

$$W = \mathcal{W}(I_{v_i}, I_{v_j}), \forall v_i, v_j \in V \quad (4)$$

where W is the weight matrix that represents weight of edges.

Then the formal definition of team network is:

Definition 5. [Team Network] A team network is constructed as a undirected weighted network $G(V, E, \mathcal{W})$ given a n -player set $V = \{v_1, v_2, \dots, v_n\}$ with l -dimensional information $I_i = \{I_1, I_2, \dots, I_l\}$, $I_i \equiv v_i \in V, \forall i = 1, 2, \dots, n$.

2.2 Network Efficiency

Let $G(V, E, W)$ be a N -player team defined in (5), a shortest path is minimum sum of weights on G between between two nodes. Then average network efficiency is

Definition 6 (Average Efficiency).

$$E(G) = \frac{1}{N(N-1)} \sum_{i \neq j \in G} \frac{1}{d_{i,j}} \quad (5)$$

where $d_{i,j}$ is the shortest path on G between $v_i, v_j \in V$.

A graph G is connected when there is a path between every pair of vertices. Notice that if there is no path between v_i and v_j , the definition $d_{i,j}$ is not defined in (6). In other words, when G is not connected, network efficiency can not be measured by (6). A component $\mathcal{G}$ of an undirected graph G is a subgraph in which any two nodes are connected to each other. Then we refer network efficiency to the following definition

Definition 7 (Network Efficiency).

$$NE(G) = \frac{1}{N} \sum_{i \in G} E(\mathcal{G}_i) \quad (6)$$

where $\mathcal{G}_i$ is a component of G .

3 Empirical Analysis

3.1 Team Network Construction on Basketball Teams

The United States National Basketball Association (NBA) was founded in 1946. It is well known as the high level of athletes. In recent years, it starts to provide data statistics from player's statistics, game logs to team performance so that makes data mining available as part of modern sport science. The data in our research is obtained from the site NBA.com/Stats, which is official database provided by NBA. Among the data, it is available information on 3736 players and 97 teams, season by season or by career. Characterization of players and teams by their statistics is playing an important role in team decision making, wage estimation and even betting market.

In this research, the data is collected from NBA.com/Stats. We select variables from 4 aspects: total played time as game played times minutes per game ($GP \times MIN$), offensive as points and assists (PTS, AST), defensive as rebounds (REB) and other information as average speed (SPD). Then we have a player node as

Definition 8 (Player Node).

$$v_i \equiv \{GP_i, MIN_i, PTS_i, AST_i, REB_i, SPD_i\}, \forall i = 1, 2, \dots, n. \quad (7)$$

where v_i represents a player and n is the total number of players over all teams.

Then we benchmark each of the player's information ONLY WITHIN their team comparing it with their team members. We refer it to internal benchmarked information. The purpose of such benchmarking is to figure out the relation between relative performance among individual team members to the overall team performance. By internal benchmarked information, the model still works when the information of other teams is not known. In practice, it could provide decision support to coaches and managers from training data in team management.

Definition 9 (Internal Benchmarked Information).

$$x_i = \frac{I_i - \min(I)}{\max(I) - \min(I)}, \text{ where } x_i \in X \quad (8)$$

where I is the information set of the team, X is the benchmarked information set of a team.

Further, we define the utility score as the average of their performance.

Definition 10. [Utility Score for Basketball Team]

$$s_i = \frac{1}{l} \sum_{j=1,2,\dots,l} x_l \quad (9)$$

where s_i is the utility score of player $v_i \in V$, $i = 1, 2, \dots, n$ and n is the number of players in a team. l is the number of dimensions for a player's information. In this case, $l = 6$.

In practice, definition (2) may not generate a integer degree sequence. So we use an adjusted function from utility score to certain all degrees are integers.

Definition 11 (Degree (Adjusted)).

$$D(v_i) = \lceil n \frac{s_i^\lambda}{\sum_i s_i^\lambda} \rceil \quad (10)$$

n is the number of players and it's the maximum edges of a node. The term $\frac{s_i^\lambda}{\sum_i s_i^\lambda}$ is actually benchmark its value between $[0, 1]$. $\lceil \cdot \rceil$ is the ceiling function.

Then given $D(v_i), v_i \in V$, we run Havel-Hakimi algorithm to generate the unweighted team network structure $G(V, E)$. Then the weight function between two players is defined here as the Euclidean distance.

Definition 12 (Weight Function for Players).

$$d_{i,j} = W(I_i, I_j) = \sqrt{\sum_{i,j=1,2,\dots,l} (I_i - I_j)^2} \quad (11)$$

where I_i is the information vector $(GP_i, MIN_i, PTS_i, AST_i, REB_i, SPD_i)$.

3.2 Team Network Structure Analysis: a Case for Cleveland Cavaliers

Insert Figure Cavalier

From the team network for Cleveland Cavaliers (2015-2016), we can see the largest component is mainly consisted of starting lineups while small components are bench players. It is interesting to see a double triangle structure among LeBron James (Forward), Kyrie Irving (Guard), Kevin Love (Forward) and Tristan Thompson (Centre-Forward). Such structure reveals the fact that the 4 players play key roles in the team.

The degree of LeBron, Kyrie and Kevin is 3 which implies they are the most organizers in both offence and defence from our information set. If one with high degree extremely underperforms in a team, its negative impact on team performance will be significant compared to the other players.

However, in figure (Warriors), Stephen Curry has a degree of 5 which means he plays a dominant role in the Golden State Warriors (GSW). From a network view, though Warriors has the highest winning probability in regular season, such network structure suffers from high risk of error, which is the extreme underperformance or injury of Stephen Curry.

Regardless the largest component as first lineup, the other components are related to bench depth. Cleveland Cavaliers (Win 69.5%) and Golden State Warriors (Win 89%) have 5 components in their team networks and there are 8-10 players in the components excluding the largest one. However, Los Angeles Lakers (Win 20.7%) only has 3 components and only 4 players outside largest component. Though Lakers also has a super star like Kobe Bryant, the lack of bench depth is a lethal wound to them.

Insert Figure Lakers

3.3 Team Performance and Network Efficiency

By the definitions above, we can have a weighted team network $G(V, E, \mathcal{W})$. Then network efficiency is calculated based on definition (7). We use linear regression to discover the relation between network efficiency and winning probability. Notice that we times the probability by 100 for ease of regression.

$$Prob_{win} = aNE(G) + b \quad (12)$$

Insert Figure

It is shown in Figure() that network efficiency is positively correlated with winning probability. The correlation is around 0.5 (for different λ). Thus, we can conclude that the network efficiency is

Insert Table

Remind that the only parameter in calculation of network efficiency (7) is λ . It is found that $\lambda \in [2.2, 2.4]$ is the proper value and the best one for fitting is 2.35. The range is in accordance with Onnela et al. (2007).

4 Conclusion

References